

Analysis 2 semester recap (multivariate real analysis)

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color code: Week ranges, Theorems, Observations/Definitions/Proofs/anything else that I deem important

W1-W3:

Def.: for $x, y \in \mathbb{R}^n$, the (standard) scalar product is defined as $x \cdot y = [x, y] = \langle x, y \rangle := \sum_{i=1}^n x_i y_i$

the euclidean norm is defined as $\|x\| := \sqrt{\sum_{i=1}^n x_i^2}$, the euclidean distance between x and y is defined as $\|x - y\| = d(x, y) := \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$

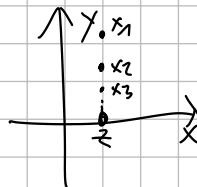
Obs.: for $n=1$, we get the abs. value of $x-y$ as the distance

Prop. (Triangle inequality): for all $x, y, z \in \mathbb{R}^n$: $\|x - z\| \leq \|x - y\| + \|y - z\|$

Proof: if you wrote LinAlg in HS/US, you probably did it already!

Def.: A sequence $(x_n)_{n \in \mathbb{N}}$ or $(x_n)_{n \in \mathbb{N}_0}$ in $\mathbb{R}^n$ is a map/function $\mathbb{N} \rightarrow \mathbb{R}^n / \mathbb{N}_0 \rightarrow \mathbb{R}^n$, and the image $x(n) = x_n$ is the sequence's n -th element

Notation: $x_s = \begin{pmatrix} x_{1,s} \\ x_{2,s} \\ \vdots \\ x_{n,s} \end{pmatrix} \in \mathbb{R}^n$ in $\mathbb{R}^2$: $x_s = \begin{pmatrix} 1 \\ 1/s \end{pmatrix}, z = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$:



Def.: Let $(x_n)_{n \in \mathbb{N}_0}$ be a sequence in $\mathbb{R}^n$. This sequence ^(seq.)converges ^(conv.) to $y \in \mathbb{R}^n$ if $\forall \varepsilon > 0 \exists N > 0$ s.t. $\forall n > N: \|x_n - y\| < \varepsilon$
 $\rightarrow y$ is the limit, with $\lim_{n \rightarrow \infty} x_n = y$, if no such limit exists, the seq. diverges.

Lemma: Each conv. series in $\mathbb{R}^n$ has a unique limit.

Obs.: The proof is analogous to the one-dimensional case (should be on my Analysis 1 recap)

Lemma: A seq. in $(\mathbb{R}^n, d(x, y) = \|x - y\|)$ convs. if and only if it converges coordinate-wise.

w.r.t. the proof:

" $\Rightarrow$ ": we know that $\|x_n - y\| \rightarrow 0$. consider the k -th coordinate: $|x_{k,n} - y_k| \leq (\sum_{i=1}^n |x_{i,n} - y_i|^2)^{1/2} = \|x_n - y\| \xrightarrow{n \rightarrow \infty} 0$

" $\Leftarrow$ ": Consider $\frac{\varepsilon}{n} > 0$ for any $\varepsilon > 0$. Then $\exists$ some index N , s.t.: $|x_{k,n} - y_k| < \frac{\varepsilon}{n} \forall n > N, \forall k \in \{1, \dots, n\}$.

Then: $\|x_n - y\| = (\sum_{i=1}^n |x_{i,n} - y_i|^2)^{1/2} \leq (\sum_{i=1}^n (\frac{\varepsilon}{n})^2)^{1/2} = (n \cdot \frac{\varepsilon^2}{n^2})^{1/2} = \sqrt{n} \cdot \frac{\varepsilon}{n} < \varepsilon$.

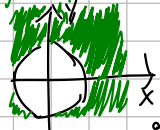
Def.: Let $x \in \mathbb{R}^n$ and $r \in \mathbb{R} > 0$. The open ball around point x with radius r is the set $B_r(x) = B(x, r) := \{y \in \mathbb{R}^n \mid \|x - y\| < r\}$

Def.: A subset $A \subset \mathbb{R}^n$ is open if for all $y \in A$, $\exists$ a ball $B_r(y)$ with $B_r(y) \subset A$. A subset $A \subset X$ is closed if $X \setminus A$ is open.

Examples (from my exercise):

Consider the set $A = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 > 1\} \rightarrow$

green: points in A



$\rightarrow$ algebraically, for the ball $B_r(y)$ of any $y \in A$, there is always some finite distance d from y to the "boundaries" of $A \rightarrow$ take $r = \frac{d}{2}$ for $B_r(y)$
 $\Rightarrow$ this set is open (easy argument: are the set bounds in the set? if it were $x^2 + y^2 \geq 1$, it would be closed)

$\Rightarrow$ convergence: $\mathbb{R} \setminus A$ is closed

$\rightarrow$ still, openness and closedness are not mutually exclusive

$\rightarrow$ consider $B = \{(x, y) \in \mathbb{R}^2 : x > 0, y \geq 0\} \rightarrow$ graphically:



$\rightarrow$ neither open nor closed (why not open? If you pick $y = (z, 0)$ with any $z > 0$, the ball $B_r(y)$ contains points outside of B too)
(why not closed? analogous reasoning for $\mathbb{R} \setminus B$)

$\rightarrow$ only two open & closed (the actual word is clopen!)

sets are $\mathbb{R}^n, \emptyset$, there are no other clopen sets in this course

Prop.: Finite intersections and infinite unions of open sets in $\mathbb{R}^n$ are open, infinite intersections and finite unions of closed sets are closed

Def.: Let $A \subset \mathbb{R}^n$ be a subset. The **interior** of a set A is defined as $A^\circ = \text{int}(A) := \bigcup \{E \subset A \mid E \text{ open}\}$ and is the largest open set included in A . The **closure** of A is defined as $\bar{A} := \bigcap \{U \supset A \mid U \text{ closed}\}$ and is the smallest closed set included in A .

The **exterior** of A is $\bar{A} \setminus A^\circ$.

Prop.: A subset $A \subset \mathbb{R}^n$ is open if and only if for all conv. seq. $(x_n)_{n \in \mathbb{N}_0}$ with limit $x \in A$, $\exists N \in \mathbb{N} \forall n > N \ x_n \in A$.

A subset $A \subset \mathbb{R}^n$ is closed if and only if for all conv. seq. $(x_n)_{n \in \mathbb{N}_0} \subset A$ with limit A : $x \in A$.

Example: $A = \{(x, y) \in \mathbb{R}^2; x \in \mathbb{Q}\}$ open? $x_n = (\frac{\sqrt{2}}{n}, 0) \xrightarrow{n \rightarrow \infty} (0, 0)$, which is in A , so A is not open
closed? we want $(x_n) \in A$, but $\lim (x_n) \notin A \rightarrow \lim x_n (\sqrt{2}, 0) \rightarrow$ define (x_n) with $x_1=1, x_2=1.4, x_3=1.41, \dots$
(you can also do this with $e, \pi, \dots$)

$\rightarrow$ **Pattern:** use the sequence propositions to find counterexamples rather than universal proofs

Rigorously proving seq. conv.: prove that $\lim_{n \rightarrow \infty} (\frac{1}{n}, \frac{1}{n^2}, s) = (0, 0, s)$: fix $\varepsilon > 0$, look for $N \in \mathbb{N}$ s.t. $\forall n > N: \|(\frac{1}{n}, \frac{1}{n^2}, s) - (0, 0, s)\| < \varepsilon$

$$\stackrel{\text{def. norm}}{\Rightarrow} \|(\frac{1}{n}, \frac{1}{n^2}, 0)\| < \varepsilon \stackrel{\circ}{\Rightarrow} \sqrt{\frac{1}{n^2} + \frac{1}{n^4}} = \frac{1}{n} \sqrt{1 + \frac{1}{n^2}} \leq \sqrt{\frac{1}{n^2} + \frac{1}{n^2}} = \frac{\sqrt{2}}{n} < \varepsilon \Rightarrow \text{choose } N = \lceil \frac{\sqrt{2}}{\varepsilon} \rceil$$

Consider $M := \{x \in \mathbb{R}^2 \mid \|x\| \leq 1\} \setminus \{0, 0\} \rightarrow \text{int}(M) = \{x \in \mathbb{R}^2 \mid 0 < \|x\| < 1\}$, $M^\circ = \{x \in \mathbb{R}^2 \mid \|x\| < 1\}$ (example)

$\rightarrow M$ not open because $B_\varepsilon(0)$ cuts M and M^c not closed either as $0 \in M^c \rightarrow$ no open ball $B_\varepsilon(0)$ lies fully in M^c

$$\rightarrow \bar{M} \setminus M^\circ = \{x \in \mathbb{R}^2 \mid \|x\| = 1\} \cup \{0\}$$

Def.: A **function** maps all allowed inputs from $\mathbb{R}^n$ to an output from the **range** ($\mathbb{R}$ or $\mathbb{R}^m$). $\mathbb{I} \rightarrow$ **domain**, $\mathbb{I} \mathbb{N}$ or $\text{image}(f) \rightarrow$ **image**,

$$f: \text{domain} \rightarrow \text{range}(f), x \mapsto y = f(x), \text{visualisation: } x \mapsto f(x) = \begin{pmatrix} f_1(x_1, \dots, x_n) \\ \vdots \\ f_n(x_1, \dots, x_n) \end{pmatrix}$$

Def.: Let $U \subset \mathbb{R}^n$ be open, $f: U \rightarrow \mathbb{R}^m$, $x^* \in U$. If for all sequences $(x_k)_{k \in \mathbb{N}_0} \subset U$ with $x_k \rightarrow x^*$, there exists a shared $y^* \in \mathbb{R}^m$ s.t. $f(x_k) \rightarrow y^*$, y^* is the **limit** of f for x_k against x^* : $\lim_{x_k \rightarrow x^*} f(x_k) = y^*$.

Def.: Let $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$. f is **continuous** if $\forall x \in X: \forall \varepsilon > 0 \exists \delta > 0$ s.t. $(\|x - x'\| < \delta \Rightarrow \|f(x) - f(x')\| < \varepsilon)$ or if for all sequences $(x_k)_{k \in \mathbb{N}_0}$ $\subset \mathbb{R}^n$ with $x_k \rightarrow x \in \mathbb{R}^n$, $(f(x_k))_{k \in \mathbb{N}_0} \subset \mathbb{R}^m$ converges against $f(x)$.

Def.: A subset $K \subset \mathbb{R}^n$ is **compact** if all sequences $(x_s)_{s \in \mathbb{N}_0} \subset K$ in K have a subseq. converging in K .

Theorem: A subset $K \subset \mathbb{R}^n$ is compact if it is closed or bounded

Corollary: let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be a cont. func. and $K \subset \mathbb{R}^n$ be compact $\Rightarrow f$ has a maximum and minimum: $\exists x \in K, x \in K$ with $f(x) = \max f(x) \leq f(x) \leq \min f(x) = f(x)$ $\forall x \in K$